Topic to discuss

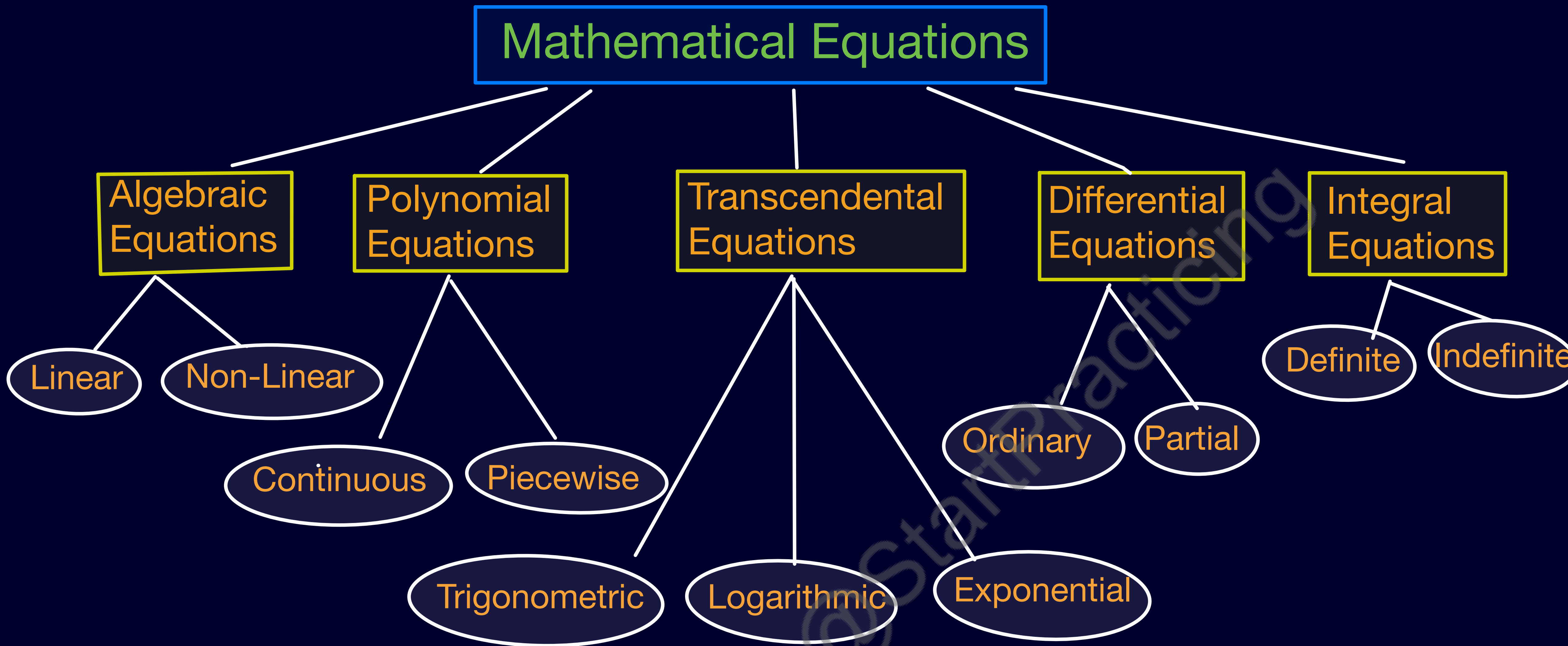
- → What is differential Equation
- → what is ordinary differential Equation
- → ODE and PDE
- → Order and Degree
- → Dependent & Independent Variable
- → Methods to solve ODE

Equation :

An equation is a mathematical statement that shows the equality between two expressions using the symbol '='

- LHS = RHS

eg: $2x + 8 = 18$



Different forms of Mathematical Equation.

Ordinary differential Equation :

An ordinary differential equation is a differential equation that involves derivatives with respect to only one independent variable.

eg: $$\frac{dy}{dx} + y = x$$

Here derivative is only with repect to x.

Partial Differential Equation :

A partial differential Equation is a differential equation that involves partial derivatives of a function with respect to two or more independent variables.

eg: $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$

- Here function depends on x and y

Order and Degree of Differential Equation :

Order : The order of a differential Equation is the heighest order derivative present

eg : $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 2$ has Order = 2

Degree : The degree is the power of the heighest order derivative (when equation is polynomial in derivatives).

eg : $\left(\frac{d^2y}{dx^2}\right)^2 + y = 2$

has degree = 2

Dependent and Independent Variable :

Independent Variable :

In differential Equations, the variable with respect to which differention is done is called independent Variable

Dependent Variable :

The variable that is being differentiated is called the dependent Variable

eg:- $\dfrac{dy}{dx} + y = x$

$x \longrightarrow$ Independent Variable

$y \longrightarrow$ Dependent Variable

Methods to solve ODE :

The solution of ODE can be obtained by two types of methods :

1) Analytical Methods → Exact solution
2) Numerical Methods → Approximate solution

Analytical Method :

→ Variable seperable Method
→ Linear differential equation method
→ Homogeneous equation
→ Exact equation

Numerical Methods :

1) Taylor Series Methods
2) Picard's Method (Successive Approximation Method)
3) Euler's Method
4) Modified Euler Method / Heun's Method
5) Runge-Kutta Method
6) Milne's Predictor Corrector Method
7) Adams-Bashforth Method

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